

BLG 317E - DATABASE SYSTEMS

FALL 2024 (CRN: 13591)

PROJECT PROPOSAL

Note that your selected dataset should be extensive enough to allow at least 5 main tables, and each table must have at least 3 non-key columns. Each team member must be responsible for at least one main table. Do your best in identifying table requirements. You will be asked to change your selected dataset if it does not seem to meet the minimum requirements. Please see the project evaluation and project self-assessment forms before finalizing your dataset to get an idea about the project evaluation criteria and process.

Project Group Name	Medico
Dataset title	Hospital Management Dataset
Dataset URL	https://www.kaggle.com/datasets/rajkumarpadmanabhan/ca-hospital-dataset-q1-2025
Data description	1) patients.csv — Patient demographics & enrollment Columns: patient_id, first_name, last_name, dob, age, gender, ethnicity, insurance_type, marital_status, address, city, state, zip, phone, email, registration_date. Represents the FHIR Patient concept; demographics are stable reference data joined by patient_id. 2) providers.csv — Clinicians & identifiers Columns: provider_id, name, department, specialty, npa, inhouse, location, years_experience, contact_info, email. npa is the U.S. National Provider Identifier (10 digits), the standard identifier in HIPAA transactions. 3) encounters.csv — Visits/episodes of care Columns: encounter_id, patient_id, provider_id, visit_date, visit_type, department, reason_for_visit, diagnosis_code, admission_type, discharge_date, length_of_stay, status, readmitted_flag. A FHIR Encounter: the actual interaction between a patient and provider(s). Use patient_id/provider_id to link. 4) diagnoses.csv — Diagnostic coding per encounter Columns: diagnosis_id, encounter_id, diagnosis_code, diagnosis_description, primary_flag, chronic_flag.

diagnosis_code is intended for ICD-10/ICD-10-CM coding of conditions; primary_flag denotes the principal diagnosis.

5) procedures.csv — Procedures/services rendered

Columns: procedure_id, encounter_id, procedure_code, procedure_description, procedure_date, provider_id, procedure_cost.

procedure_code is typically CPT (or HCPCS) for billing/reporting of services.

6) lab_tests.csv — Laboratory tests & results

Columns: lab_id, encounter_id, test_name, test_code, specimen_type, test_result, units, normal_range, test_date, status.

test_code is designed for LOINC, the standard coding for lab orders/results; each record is an individual lab observation.

7) medications.csv — Medications prescribed/administered

Columns: medication_id, encounter_id, drug_name, dosage, route, frequency, duration, prescribed_date, prescriber_id, cost.

Captures medication orders/administrations tied to an encounter (maps to FHIR medication resources conceptually).

8) claims_and_billings.csv — Charges, claims, payments

Columns: billing_id, patient_id, encounter_id, insurance_provider, payment_method, claim_id, claim_billing_date, billed_amount, paid_amount, claim_status, denial_reason.

Represents charge/claim lines and adjudication status at a summary level; claim_id links to denial details. The FHIR Claim resource is the analog for reimbursement requests.

9) denials.csv — Denials & appeals lifecycle

Columns: claim_id, denial_id, denial_reason_code, denial_reason_description, denied_amount, denial_date, appeal_filed, appeal_status, appeal_resolution_date, final_outcome.

Tracks denial reasons and appeal outcomes per claim. Useful for payer performance and revenue cycle analysis. (Claim/denial processes align with FHIR financial resources.)

Member 1
Signature

Full name	Abdulsamet Ekinci
Student no	150220723
Responsibility:	Managing team – DB Analysis – patients -

		encounters
Member 2 Signature	Full name	Uğur Erkan Bol
	Student no	150210074
	Responsibility	UI/UX design – Tester – providers - diagnoses
Member 3 Signature	Full name	Furkan İslamoğlu
	Student no	150220065
	Responsibility:	Frontend Lead – procedures – medications
Member 3 Signature	Full name	Mustafa Talha Hamsoğlu
	Student no	150220066
	Responsibility:	Backend Lead – claims_and_billings - denials
Member 4 Signature	Full name	Mustafa Yunus Diler
	Student no	150210003
	Responsibility:	Backend Optimization – lab_tests

